

Reliable Individual Fairness Evaluation through Coverage, Metric Agreement, and Model Stability

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Abstract. Individual-fairness conclusions can depend on neighbourhood quality, similarity metrics, and model choice. We propose a validation framework that integrates these factors into a diagnostic component vector and a composite score. The framework is evaluated on four binary-classification datasets using controlled synthetic injections targeting sparse support, dominant neighbours, metric disagreement, nonlinear model boundaries, and combined failures. The targeted manipulations generally reduce the intended components, while the clean condition remains comparatively stable. Compared with three existing validation approaches based on statistical inference, formal verification, and discriminatory-instance validity, the proposed framework provides broader information about why an empirical fairness conclusion weakens. The results indicate that individual-fairness evaluation should assess not only outcome similarity, but also the strength and robustness of the evidence supporting it.

Keywords: Individual fairness · Fairness validation · Similarity metrics

1 Introduction

Machine learning systems increasingly support decisions in consequential domains such as healthcare, employment, lending, and criminal justice. Although these systems may improve the consistency and scalability of decision-making, they can also reproduce or amplify unfair patterns present in data, model design, or deployment environments. Algorithmic fairness has therefore become an important requirement for evaluating whether machine learning systems treat individuals and social groups appropriately [8]. Among the principal notions of algorithmic fairness, individual fairness requires that individuals who are similar with respect to a task-relevant similarity measure receive similar predictions or decisions [2]. Unlike group-based criteria, which compare aggregate outcomes across demographic groups, individual fairness evaluates treatment at the level of particular individuals and their relevant counterparts.

Evaluating individual fairness, however, requires more than computing the difference between model outputs for selected pairs. Any empirical conclusion about individual fairness depends on several analytical choices. First, an individual must have sufficiently

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similar observations against which their treatment can be meaningfully compared. A fixed nearest-neighbour procedure will always return neighbours, even when the closest available observations are distant or when almost all comparison weight is concentrated on a single neighbour. In such cases, a numerical fairness score may be computable but supported by weak local evidence. Second, the identities and similarities of the selected neighbours depend on the distance metric used to represent task-relevant similarity. Euclidean, Manhattan, cosine, learned, or domain-specific metrics can induce substantially different neighbourhood structures. Consequently, a fairness conclusion obtained under one metric may not remain credible when other plausible similarity definitions are considered. Third, fairness may differ across predictive models trained on the same data. A conclusion derived from one model may therefore reflect model-specific decision boundaries rather than a stable pattern in the learning problem.

These dependencies create an important distinction between measuring individual fairness and validating an individual-fairness conclusion. Existing work has developed methods for statistically auditing individual fairness [14], identifying discriminatory instances through fairness testing [10], and formally verifying whether specified fairness properties hold for particular classes of models [3, 1]. Other approaches examine whether generated discriminatory instances are realistic enough to support testing and retraining [4], or certify individual fairness under local and distributional perturbations [11]. These methods address important questions concerning violation detection, statistical inference, input validity, and formal guarantees. However, comparatively less attention has been given to whether an empirical fairness result is jointly supported by adequate neighbourhood evidence, agreement across plausible similarity measures, favourable individual-level outcomes, and stability across predictive models.

We propose a four-component framework evaluating neighbourhood coverage, distance metric reliability, individual fairness, and model stability. The components are reported diagnostically and combined into a composite validation score. Using controlled synthetic injections on real tabular datasets, we test whether each component responds to its targeted weakness and compare the framework with three existing validation approaches. Our contributions are: (i) distinguishing fairness measurement from validation of its supporting evidence; (ii) introducing a four-component diagnostic framework; and (iii) developing a severity-controlled injection protocol for evaluating diagnostic sensitivity.

2 Related Work

Individual fairness generally requires similar individuals to receive similar outcomes under a task-relevant similarity measure [2]. Existing validation approaches address statistical inference, formal verification, and the validity of generated discriminatory instances. We compare against one representative framework from each category.

VF1: Statistical inference for individual fairness. Maity et al. [7] assess individual fairness through an adversarial loss ratio. Given a model f , loss function ℓ , fair metric d_x , and an unfair map $\Phi(x, y)$ that searches for a nearby instance with larger model loss, the population audit value is

$$A(f) = \mathbb{E}_{(x,y) \sim P} \left[\frac{\ell(f(\Phi(x,y)), y)}{\ell(f(x), y)} \right]. \quad (1)$$

For an audit sample of size n , it is estimated by

$$S_n = \frac{1}{n} \sum_{i=1}^n \frac{\ell(f(\Phi(x_i, y_i)), y_i)}{\ell(f(x_i), y_i)}, \quad (2)$$

with sample standard deviation V_n . The one-sided lower confidence statistic used for testing is

$$T_n = S_n - \frac{z_{1-\alpha}}{\sqrt{n}} V_n. \quad (3)$$

The null hypothesis $H_0 : A(f) \leq \delta$ is rejected when $T_n > \delta$. Thus, larger values indicate stronger statistical evidence of unfairness. VF1 provides inferential calibration for a specified model and fair metric, but does not separately measure neighbourhood support, agreement among several metrics, or stability across model classes.

VF2: Formal verification of individual fairness. George John et al. [3] define a model as biased when there exists a pair of valid inputs that are sufficiently close but receive substantially different outputs. In general form, VF2 searches for

$$\exists x, x' \in X_{\text{valid}} \quad \text{such that} \quad d_x(x, x') \leq \varepsilon \quad \wedge \quad d_y(f(x), f(x')) > \delta, \quad (4)$$

where ε specifies permissible input variation and δ is the maximum acceptable output difference. Equivalently, verification may be written as the constrained maximisation

$$\max_{\substack{x, x' \in X_{\text{valid}} \\ d_x(x, x') \leq \varepsilon}} d_y(f(x), f(x')). \quad (5)$$

A value exceeding δ provides a counterexample to individual fairness; otherwise, the verifier attempts to certify that no such pair exists under the specified model and domain constraints. In our experiments, its empirical output is summarised as the proportion of checked pairs satisfying the violation condition,

$$R_{\text{VF2}} = \frac{\sum_{(x, x') \in \mathcal{P}} \mathbb{I}[d_x(x, x') \leq \varepsilon \wedge d_y(f(x), f(x')) > \delta]}{|\mathcal{P}|}. \quad (6)$$

This rate is our experimental summary of VF2 behaviour rather than the original paper's formal certification output.

VF3: Individual-fairness testing with data validity. Kitamura et al. [4] extend individual-fairness testing by requiring detected discriminatory instances to satisfy a data-validity constraint. Let $\mathcal{G}$ be the set of generated discriminatory instances, where an instance $x' \in \mathcal{G}$ satisfies the testing criterion

$$d_x(x, x') \leq \varepsilon \quad \wedge \quad d_y(f(x), f(x')) > \delta. \quad (7)$$

Let $v(x') \in \{0, 1\}$ indicate whether x' satisfies the learned validity constraints, including the permitted feature interactions. The valid discriminatory-instance rate is

$$R_{\text{VF3}} = \frac{\sum_{x' \in \mathcal{G}} v(x')}{|\mathcal{G}|}. \quad (8)$$

A larger value indicates that a greater proportion of the generated discriminatory instances are plausible according to the validity model. VF3 therefore evaluates the credibility of generated test cases, rather than the coverage, cross-metric agreement, or cross-model stability of the underlying fairness conclusion.

These frameworks answer complementary questions. VF1 quantifies statistical evidence against a fairness hypothesis, VF2 searches for violations of a formally specified property, and VF3 measures the validity of generated discriminatory instances. The proposed framework instead assesses whether an empirical fairness conclusion is jointly supported by neighbourhood coverage, distance-metric reliability, outcome similarity, and model stability.

3 Proposed Validation Framework

3.1 Overview and Notation

Let $X = \{x_i\}_{i=1}^n$ denote a dataset of n individuals. Indices i and j denote a focal individual and one of its neighbours, respectively; q indexes distance metrics, and ℓ indexes predictive models. Let $\mathcal{D} = \{d^{(1)}, \dots, d^{(Q)}\}$ denote the selected distance metrics and $\mathcal{F} = \{f_1, \dots, f_L\}$ the trained models. For binary classification, $p_i^{(\ell)} = f_\ell(x_i) \in [0, 1]$ is the probability predicted for individual i by model ℓ .

The framework evaluates four dimensions: neighbourhood coverage C , distance-metric reliability D , individual fairness F , and model stability M . Each lies in $[0, 1]$, with higher values indicating stronger support for the fairness conclusion.

3.2 Neighbourhood Construction and Coverage

For each focal individual i , let $\mathcal{N}_i$ contain its k nearest neighbours under Euclidean distance. Each neighbour $j \in \mathcal{N}_i$ receives the Gaussian weight

$$w_{ij}^{(E)} = \exp\left(-\frac{(d_{ij}^{(E)})^2}{2(h_i^{(E)})^2}\right), \quad h_i^{(E)} = \max\left\{\text{median}_{j \in \mathcal{N}_i} d_{ij}^{(E)}, \varepsilon\right\} \quad (9)$$

where $d_{ij}^{(E)}$ is the Euclidean distance between individuals i and j , and $\varepsilon > 0$ prevents division by zero. The Gaussian kernel converts distance into a smooth, bounded similarity weight, so closer neighbours contribute more strongly without introducing the discontinuity of a hard threshold. The bandwidth is defined using the median distance within each local neighbourhood, which scales similarity relative to the typical neighbour distance around each individual and is less sensitive to unusually close or distant observations.

Coverage accounts for both the distribution and absolute strength of the neighbour weights:

$$\text{ESS}_i = \frac{\left(\sum_{j \in \mathcal{N}_i} w_{ij}^{(E)}\right)^2}{\sum_{j \in \mathcal{N}_i} (w_{ij}^{(E)})^2}, \quad c_i = \frac{\text{ESS}_i}{k} \left(\frac{1}{k} \sum_{j \in \mathcal{N}_i} w_{ij}^{(E)} \right). \quad (10)$$

The dataset-level coverage score is $C = \frac{1}{n} \sum_{i=1}^n c_i$. It is high when each individual has several sufficiently similar neighbours rather than one dominant or uniformly distant comparison. Effective Sample Size (ESS) is used because the nominal value of k can overstate the amount of available comparison evidence when most of the total weight is concentrated on only one or two neighbours. The normalised term ESS_i/k measures how evenly the evidence is distributed, while the mean neighbour weight measures whether the retained neighbours are genuinely close. Their product prevents uniformly weak neighbours from receiving high coverage merely because their weights are evenly distributed.

3.3 Distance-Metric Reliability

Each metric $d^{(q)} \in \mathcal{D}$ is evaluated on the same retained pairs. Its distance is converted into a similarity weight:

$$s_{ij}^{(q)} = \exp\left(-\frac{(d_{ij}^{(q)})^2}{2(h_i^{(q)})^2}\right), \quad h_i^{(q)} = \max\left\{\text{median}_{j \in \mathcal{N}_i} d_{ij}^{(q)}, \varepsilon\right\}. \quad (11)$$

For pair (i, j) , let $\bar{s}_{ij}$ be the mean similarity across the Q metrics and σ_{ij} their population standard deviation:

$$\bar{s}_{ij} = \frac{1}{Q} \sum_{q=1}^Q s_{ij}^{(q)}, \quad \sigma_{ij} = \sqrt{\frac{1}{Q} \sum_{q=1}^Q \left(s_{ij}^{(q)} - \bar{s}_{ij}\right)^2}. \quad (12)$$

Pair-level reliability is $r_{ij} = \max\{0, 1 - 2\sigma_{ij}\}$. Standard deviation is used because it directly measures disagreement among the metric-specific similarity values and equals zero under complete agreement. Since all similarities lie in $[0, 1]$, their population standard deviation is at most 0.5. The transformation $1 - 2\sigma_{ij}$ therefore yields a bounded reliability measure that equals one under complete agreement and decreases toward zero as disagreement increases. The dataset-level reliability score is

$$D = \frac{\sum_{i=1}^n \sum_{j \in \mathcal{N}_i} \bar{s}_{ij} r_{ij}}{\sum_{i=1}^n \sum_{j \in \mathcal{N}_i} \bar{s}_{ij}}. \quad (13)$$

The reliability values are weighted by $\bar{s}_{ij}$ so that disagreement about pairs regarded as strongly similar has greater influence than disagreement about pairs that all metrics already consider weak comparisons. Thus, D is high when the selected metrics assign similar similarity values to the retained pairs.

3.4 Individual Fairness and Model Stability

The consensus weight for pair (i, j) is $\omega_{ij} = \bar{s}_{ij}r_{ij}$. This product requires a pair to be both strongly similar on average and consistently judged similar across metrics before it receives substantial influence in the fairness calculation. For model ℓ , the weighted outcome discrepancy and fairness score are

$$\Delta_\ell = \frac{\sum_{i=1}^n \sum_{j \in \mathcal{N}_i} \omega_{ij} |p_i^{(\ell)} - p_j^{(\ell)}|}{\sum_{i=1}^n \sum_{j \in \mathcal{N}_i} \omega_{ij}}, \quad F_\ell = 1 - \Delta_\ell. \quad (14)$$

Absolute probability difference is used because it is symmetric, bounded, and measures the magnitude of unequal treatment without allowing positive and negative differences to cancel. Subtracting the discrepancy from one gives the fairness component the same direction as the other components, with higher values indicating stronger support.

The overall fairness component is $F = \frac{1}{L} \sum_{\ell=1}^L F_\ell$. Model stability is calculated from the variation in these model-specific scores:

$$\sigma_F = \sqrt{\frac{1}{L} \sum_{\ell=1}^L (F_\ell - F)^2}, \quad M = \max\{0, 1 - 2\sigma_F\}. \quad (15)$$

The mean fairness score F captures the typical level of individual fairness across models, whereas M captures whether that conclusion is model-dependent. The same standard-deviation transformation is used because the model-specific fairness scores are also bounded within $[0, 1]$. A high F indicates similar outcomes among similar individuals, while a high M indicates that this conclusion is consistent across models.

3.5 Composite Validation Score

The overall validation score is

$$V = C \times D \times F \times M. \quad (16)$$

Multiplication is used to express a conjunctive notion of credibility: strong performance on one dimension should not fully compensate for a serious weakness on another. For example, a high fairness score should not produce a high validation result when it is based on poor neighbourhood evidence or varies substantially across models. Since all components lie in $[0, 1]$, $V \in [0, 1]$. Hence, this form reflects the requirement that a well-supported fairness conclusion should have adequate neighbourhood evidence, agreement across distance metrics, favourable individual-level outcomes, and stability across models. Because equal values of V may reflect different weaknesses, the component vector (C, D, F, M) is always reported alongside the composite score.

4 Experimental Design

4.1 Datasets, Models, and Distance Metrics

We evaluated the proposed framework on four binary-classification datasets commonly used in individual-fairness research (Table 1). Continuous features were standardised

and categorical features were numerically encoded. For each dataset, we used a stratified train–test split, trained all predictive models on the training set, and computed fairness-related quantities on the test set.

Table 1. Datasets used in the experiments.

ID	Dataset	Instances	Features	Class counts ($Y = 0, Y = 1$)
D1	Breast Cancer Wisconsin Diagnostic [12]	569	30	(212, 357)
D2	Pima Indians Diabetes [9]	768	8	(500, 268)
D3	Blood Transfusion Service Center [15]	748	4	(570, 178)
D4	Banknote Authentication [6]	1372	4	(762, 610)

To assess stability across model classes, we used Logistic Regression, Support Vector Machine, Random Forest, and XGBoost [13]. Predicted probabilities were used in the individual-fairness calculations. Similarity was evaluated using Euclidean, Manhattan, and cosine distance [5]. Euclidean distance was used to construct the reference k -nearest-neighbour sets, while all three metrics were used when computing distance-metric reliability.

4.2 Synthetic Instances Injection

To evaluate whether the framework responds correctly to known weaknesses, we introduced synthetic instances into the test data at increasing injection rates. We used $\rho \in \{0, 0.05, 0.10, 0.20, 0.30\}$. Six scenarios were considered:

- **Clean dense clusters:** baseline condition with no targeted weakness.
- **Sparse isolated instances:** synthetic points placed away from dense regions to reduce local comparison support.
- **Dominant neighbour:** local configurations constructed so that one neighbour receives disproportionately large similarity weight.
- **Metric disagreement geometry:** instances designed so that Euclidean, Manhattan, and cosine metrics disagree about similarity.
- **Nonlinear model boundary:** instances placed near regions where linear and non-linear models produce different local decision behaviour.
- **Combined manipulations:** a mixture of the above perturbations.

These scenarios were designed to target different components of the framework. Sparse isolated instances and dominant-neighbour structures primarily weaken coverage C , metric-disagreement geometry primarily weakens distance-metric reliability D , nonlinear model boundaries primarily weaken model stability M , and the combined setting tests whether the framework reflects multiple simultaneous weaknesses.

4.3 Comparison with Existing Validation Frameworks

We compared the proposed framework against three existing validation approaches introduced in Section 2: VF1, VF2, and VF3. Because these methods produce different types of outputs, we did not force them onto a common scale. Instead, each method was evaluated using its native output:

- **Proposed framework:** composite validation score V , together with its component scores C , D , F , and M .
- **VF1:** lower confidence statistic.
- **VF2:** violation rate among checked pairs.
- **VF3:** valid discriminatory-instance rate.

5 Results and Discussion

Figure 1 presents the component scores across increasing synthetic-instance rates. The clean dense-cluster condition remained broadly stable, indicating that the framework did not penalise the addition of synthetic observations in the absence of a targeted weakness. In contrast, the manipulated conditions produced component-level changes broadly consistent with their intended effects.

Sparse isolated instances reduced coverage C from approximately 0.64 to 0.56, while the remaining components changed only slightly. The dominant-neighbour condition produced a comparable decline in C , consistent with the reduction in effective comparison support when similarity weight becomes concentrated on a small number of neighbours. Metric-disagreement geometry primarily reduced distance-metric reliability D , while model stability remained nearly unchanged. This indicates that the framework can distinguish disagreement in similarity measurement from instability in model behaviour.

The nonlinear model-boundary condition produced the strongest reduction in model stability M , accompanied by a decline in individual fairness F . This reflects increasing disagreement among model-specific fairness scores in regions where linear and nonlinear models form different decision boundaries. The combined-manipulation condition reduced several components simultaneously, particularly C , F , and M . The component vector (C, D, F, M) therefore reveals not only whether a fairness conclusion has weakened, but also the likely source of that weakness.

As shown in Fig. 2, the composite score V remained stable or increased slightly for the clean condition, whereas all targeted manipulations reduced it as injection severity increased. The nonlinear model-boundary condition produced the largest decline, followed by the combined-manipulation condition. Sparse isolated instances, dominant-neighbour structures, and metric disagreement produced smaller but consistent reductions. This behaviour supports the multiplicative design of V : a serious weakness in one component lowers the overall validation result, while simultaneous weaknesses cause a stronger decline.

The comparison methods responded to narrower subsets of the injected conditions. VF1 reacted strongly to nonlinear model-boundary manipulation, although its response was not consistently monotonic across all scenarios. VF2 showed substantial increases in

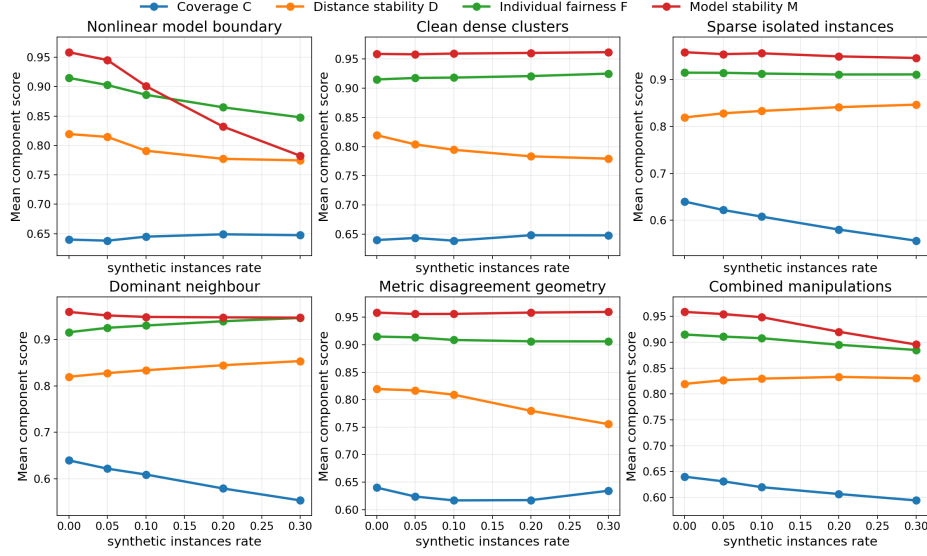


Fig. 1. Mean framework component scores for synthetic-injection scenarios and injection rates across datasets. Each manipulation targets a different source of weakness in an individual-fairness conclusion.

violation rate for the nonlinear-boundary and combined conditions but changed little under sparse, dominant-neighbour, and metric-disagreement manipulations. VF3 similarly responded most clearly when the injections generated valid discriminatory instances, while remaining comparatively insensitive to weak neighbourhood support or metric disagreement.

These differences reflect the distinct objectives of the methods. VF1 evaluates statistical evidence against a fairness hypothesis, VF2 searches for violations of a formally specified property, and VF3 evaluates whether discriminatory test instances are valid. The proposed framework instead assesses the evidential conditions surrounding an empirical fairness conclusion. Its advantage is therefore not universal numerical superiority, but broader diagnostic coverage across neighbourhood support, metric agreement, individual outcomes, and model dependence.

The multiplicative score should nevertheless be interpreted together with its components. A low value of V identifies weak joint support but does not alone reveal whether the cause is poor coverage, metric disagreement, unfair outcomes, or model instability. The diagnostic vector remains the primary explanation, while V provides a compact summary.

6 Limitations

The evaluation uses controlled synthetic manipulations whose intended effects are known in advance. This provides diagnostic ground truth but may not capture the

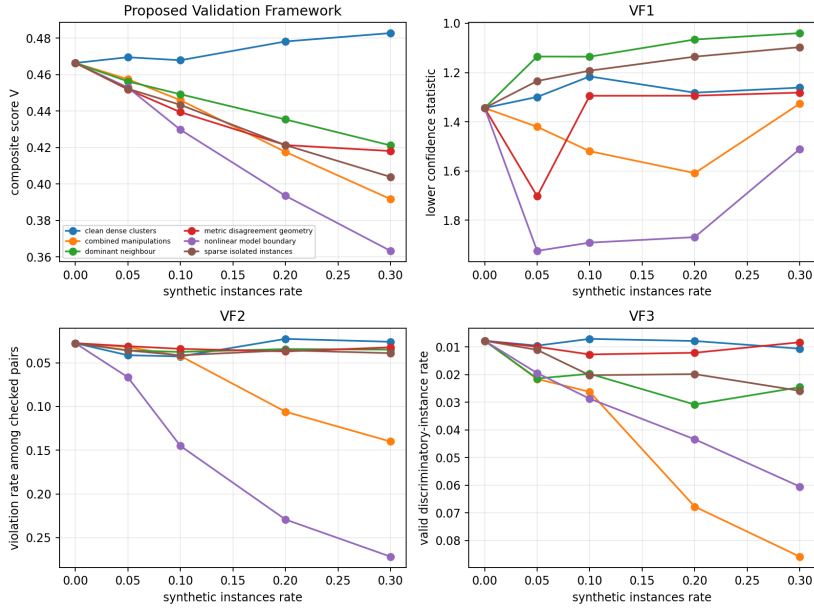


Fig. 2. Comparison of the proposed composite validation score with VF1, VF2, and VF3 for synthetic-injection scenarios across datasets. The vertical axes for VF1–VF3 are reversed because larger native values indicate worse outcomes; consequently, downward movement represents deterioration in all four panels.

complexity of unfairness arising in real deployment settings. Further evaluation under naturally occurring distribution shifts and operational decision systems is therefore needed.

The framework is also conditional on the selected feature representation, distance metrics, neighbourhood size, bandwidth rule, and predictive models. Considering multiple metrics and models reduces dependence on any single choice, but does not establish that those choices are normatively appropriate. In particular, Euclidean distance is used to construct the reference neighbourhood, which may exclude relationships considered close only under another metric.

Finally, the experiments use small and medium-sized tabular binary-classification datasets. The framework’s behaviour and computational cost on large-scale, high-dimensional, multimodal, or sequential data remain untested. The resulting score should therefore be interpreted as an empirical assessment of support under the specified analytical configuration, not as a formal or ethical certificate of fairness.

7 Conclusion

This paper introduced a multi-dimensional framework for validating empirical individual-fairness conclusions through neighbourhood coverage, distance-metric reliability, indi-

vidual fairness, and model stability. Rather than treating a fairness score as sufficient on its own, the framework evaluates whether that score is supported by meaningful local comparisons, consistent similarity relationships, comparable outcomes for similar individuals, and stability across predictive models.

Controlled synthetic injections showed that the framework responded to distinct sources of weakness in the intended components. Sparse and dominant-neighbour conditions reduced coverage, metric-disagreement geometry reduced reliability, and nonlinear decision-boundary conditions weakened fairness and model stability. The composite score summarised these effects, while the component vector retained their diagnostic interpretation. Compared with VF1, VF2, and VF3, the proposed framework provided broader information about why an empirical fairness conclusion became less credible, although the methods remain complementary rather than directly interchangeable.

Overall, the results suggest that individual-fairness evaluation should consider not only whether similar individuals receive similar outcomes, but also whether the evidence supporting that conclusion is adequate and robust. Future work should examine the framework under real deployment shifts, alternative neighbourhood constructions, larger datasets, and non-tabular learning settings.

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